

2(a)	distance (from the point) in a straight line in a given direction	B1
2(b)(i)	distance = speed $\times$ time = 6.0×0.71 = 4.3 m	A1
2(b)(ii)	$s = ut + \frac{1}{2}at^2$ = $\frac{1}{2} \times 9.81 \times 0.71^2$	C1
	= 2.5 m	A1
2(b)(iii)	$\tan \theta = 2.5 / 4.3$ or hypotenuse = $\sqrt{4.3^2 + 2.5^2}$ (= 4.97 m) $\cos \theta = 4.3 / 4.97$ or $\sin \theta = 2.5 / 4.97$	C1
	$\theta = 30^\circ$	A1

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Question	Answer	Marks
2(b)(iv)	displacement = $\sqrt{4.3^2 + 2.5^2}$	C1
	= 4.9 m or 5.0 m	A1
	or	
	displacement = $2.5 / \sin 30^\circ$ or displacement = $4.3 / \cos 30^\circ$	(C1)
	= 5.0 m	(A1)
2(b)(v)	KE = $\frac{1}{2}mv^2$ or GPE = mgh	C1
	initial KE + loss in GPE = final KE $(\frac{1}{2} \times m \times 6.0^2) + (m \times 9.81 \times 2.5) = (\frac{1}{2} \times m \times v^2)$	C1
	$v = 9.2 \text{ m s}^{-1}$	A1